



CSE3023 WEB-BASED APPLICATION DEVELOPMENT

LAB REPORT 6

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PROGRAMME: COMPUTER SCIENCE (SOFTWARE ENGINEERING)

Table of Content

User.java.....	2
UserDB.java.....	3
UserDAO.java.....	4
InsertUser.jsp.....	5
processUser.jsp.....	6
login.jsp.....	7
doLogin.jsp.....	8
main.jsp.....	9

LAB EXERCISE

```
package lab6.com;

public class User {
    private String username;
    private String password;
    private String fname;
    private String lname;

    public User(){}

    public String getUsername() {
        return username;
    }

    public void setUsername(String username) {
        this.username = username;
    }

    public String getPassword() {
        return password;
    }

    public void setPassword(String password) {
        this.password = password;
    }

    public String getFname() {
        return fname;
    }

    public void setFname(String fname) {
        this.fname = fname;
    }

    public String getLname() {
        return lname;
    }

    public void setLname(String lname) {
        this.lname = lname;
    }
}
```

User.java

```

package lab6.com;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class UserDB {
    private static Connection conn = null;
    private static String jdbcURL = "jdbc:mysql://localhost:3306/csf3107";
    private static String jdbcUser = "root";
    private static String jdbcPassword = "admin";

    public static Connection getConnection() throws ClassNotFoundException{
        try {
            if (conn != null && !conn.isClosed()) {
                return conn;
            }

            Class.forName("com.mysql.cj.jdbc.Driver");
            conn = DriverManager.getConnection(jdbcURL, jdbcUser,
jdbcPassword);

            }catch(SQLException e){
                e.printStackTrace();
            }
            return conn;
        }

        public static void closeConnection() throws ClassNotFoundException{
            try{
                if(conn != null && !conn.isClosed()){
                    conn.close();
                }
            }catch(SQLException e){
                e.printStackTrace();
            }finally{
                conn = null;
            }
        }
    }
}

```

UserDB.java

```

package lab6.com;

import java.sql.*;

public class UserDAO {
    private Connection conn;
    private int result = 0;

    public UserDAO() throws ClassNotFoundException{
        conn = UserDB.getConnection();
    }

    public int insertUser(User user){
        try {
            String sql = "INSERT INTO userprofile (username, password,
firstname, lastname) VALUES (?, ?, ?, ?)";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, user.getUsername());
            pstmt.setString(2, user.getPassword());
            pstmt.setString(3, user.getFname());
            pstmt.setString(4, user.getLname());
            result = pstmt.executeUpdate();

        } catch (SQLException e) {
            e.printStackTrace();
        }
        return result;
    }

    public User validateUser(String username, String password) throws
SQLException {
        User validUser = null;
        String sql = "SELECT * FROM userprofile WHERE username = ? AND
password = ?";

        try (PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, username);
            pstmt.setString(2, password);

            try (ResultSet rs = pstmt.executeQuery()) {
                if (rs.next()) {
                    validUser = new User();
                    validUser.setUsername(rs.getString("username"));
                    validUser.setFname(rs.getString("firstname"));
                    validUser.setLname(rs.getString("lastname"));
                }
            }
        }
        return validUser;
    }
}

```

UserDAO.java

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>User Registration</title>
</head>
<body>
<form action="processUser.jsp" method="post">
  <fieldset>
    <legend>New User Registration</legend>
    Username: <input type="text" name="username"><br><br>
    Password: <input type="password" name="password"><br><br>
    First Name: <input type="text" name="fname"><br><br>
    Last Name: <input type="text" name="lname"><br><br>
    <button type="submit">Register</button>
    <button type="reset">Cancel</button>
  </fieldset>
</form>
</body>
</html>
```

InsertUser.jsp

```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%@ page import="java.sql.*" %>
<%@ page import="lab6.com.UserDAO" %>
<%@ page import="lab6.com.UserDB" %>

<html>
<head>
    <title>Title</title>
</head>
<body>
<jsp:useBean id="user" class="lab6.com.User" scope="request"/>

<jsp:setProperty name="user" property="*" />

<%
    try {
        UserDAO userDAO = new UserDAO();
        int result = userDAO.insertUser(user);

        if (result > 0) {
            response.sendRedirect("login.jsp");
        } else {
            out.print("<p style='color:red;'>Registration failed. Please try
again.</p>");
            out.print("<a href='insertUser.html'>Try Again</a>");
        }
    } catch (Exception e) {
        e.printStackTrace();
        out.print("<p style='color:red;'>SQL Error: " + e.getMessage() +
"</p>");
        out.print("<a href='insertUser.html'>Try Again</a>");
    } finally {
        UserDB.closeConnection();
    }
%>
</body>
</html>

```

processUser.jsp

```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<html>
<head>
    <title>Title</title>
</head>
<body>
<form action="doLogin.jsp" method="post">
    <fieldset style="width: 300px;">
        <legend>User Login</legend>

        <% if(request.getParameter("msg") != null) { %>
            <p style="color: red; font-weight: bold;"><%=
request.getParameter("msg") %></p>
        <% } %>

        Username: <input type="text" name="username" required><br><br>
        Password: <input type="password" name="password" required><br><br>
        <button type="submit">Login</button>
    </fieldset>
</form>
</body>
</html>

```

login.jsp

```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%@ page import="lab6.com.User" %>
<%@ page import="lab6.com.UserDAO" %>
<%@ page import="lab6.com.UserDB" %>

<%
String usernameIn = request.getParameter("username");
String passwordIn = request.getParameter("password");

UserDAO userDAO = new UserDAO();

try {
    User validatedUser = userDAO.validateUser(usernameIn, passwordIn);

    if (validatedUser != null) {
        session.setAttribute("username", validatedUser.getUsername());
        session.setAttribute("firstname", validatedUser.getFname());
        session.setAttribute("lastname", validatedUser.getLname());

        response.sendRedirect("main.jsp");
    } else {
        response.sendRedirect("login.jsp?msg=Invalid username or
password..!");
    }
} catch (Exception e) {
    e.printStackTrace();
    response.sendRedirect("login.jsp?msg=System Database Error");
} finally {
    UserDB.closeConnection();
}
%>

```

doLogin.jsp

```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<html>
<head>
    <title>Main Dashboard</title>
</head>
<body>
<%
    if (session.getAttribute("username") == null) {
        response.sendRedirect("login.jsp?msg=Please log in first.");
    } else {
%>
<h1>Welcome to the Main Page</h1>
<table border="1" cellpadding="10" style="border-collapse: collapse;">
    <tr style="background-color: #f2f2f2;"><th colspan="2">Logged In User
Profile Details</th></tr>
    <tr><td><strong>Username:</strong></td><td><%=
session.getAttribute("username") %></td></tr>
    <tr><td><strong>First Name:</strong></td><td><%=
session.getAttribute("firstname") %></td></tr>
    <tr><td><strong>Last Name:</strong></td><td><%=
session.getAttribute("lastname") %></td></tr>
</table>
<br>
<a href="logout.jsp">Log Out Safely</a>
<% } %>
</body>
</html>

```

main.jsp

OUTPUT:

← → ↻ ⓘ localhost:8080/Lab6_war_exploded/insertUser.html

New User Registration

Username:

Password:

First Name:

Last Name:

Register User page

← → ↻ ⓘ localhost:8080/Lab6_war_exploded/login.jsp

User Login

Username:

Password:

Login page after registered successfully

← → ↻ ⓘ localhost:8080/Lab6_war_exploded/main.jsp

Welcome to the Main Page

Logged In User Profile Details	
Username:	Jojo
First Name:	Jojo
Last Name:	Yap

[Log Out Safely](#)

Main page